PROJECT DOCUMENTATION

Case Study

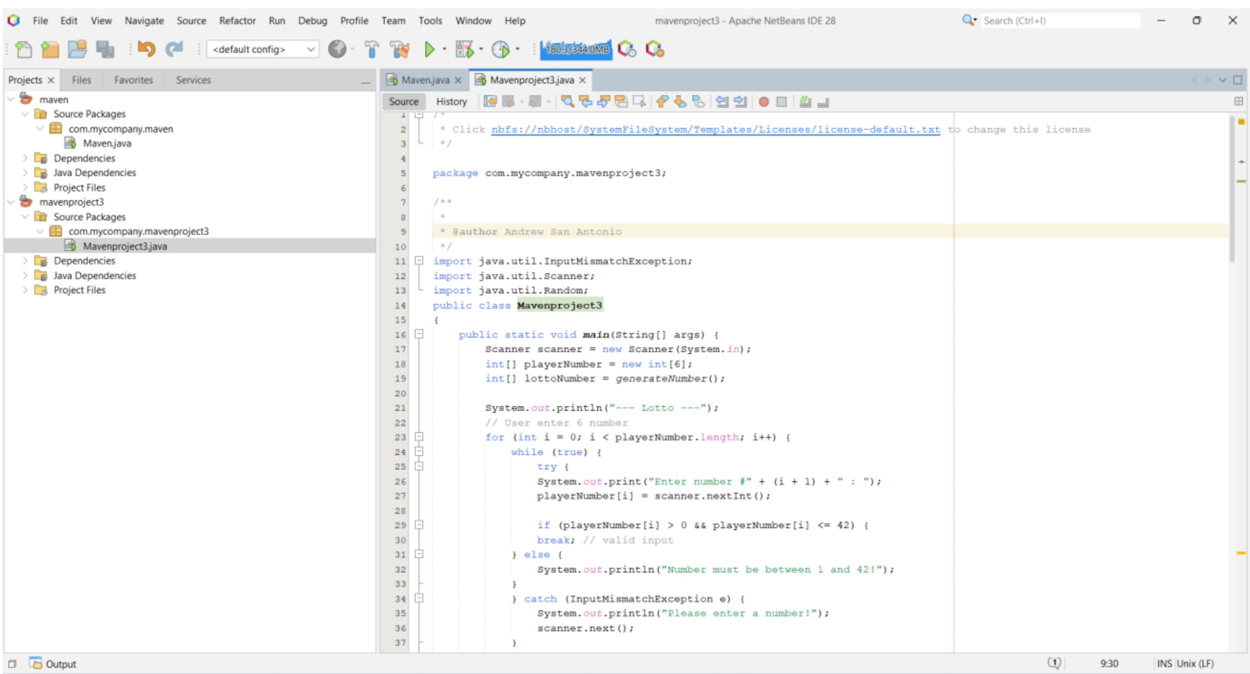
Arrays

Lotto - A lottery program made of 6 random numbers (Lotto 6/42).

Design a program that:

- Allows the user to enter 6 numbers as their bet.
- Generates the 6 random numbers for the winning combination. (Use the random function in Java.)
- Display the winning numbers to make it easier to check.
- Display the prize of the lotto buyer based on the following:

Winning Numbers (in any order)	Prize
6 winning numbers	P 31,621,223.97
5 winning numbers	P 25,000.00
4 winning numbers	P 1,000.00
3 winning numbers	P 20.00



```java
            }
        }
        // New Line
        System.out.println();

        // Display Bet Numbers and Lotto Numbers
        displayNumbers(playerNumber, lottoNumber);
        // Display Winning Numbers;
        displayWinningNumbers(playerNumber, lottoNumber);
        // Display Prize of the winners
        displayPrize(playerNumber, lottoNumber);
    }
    // Generate 6 random numbers
    public static int[] generateNumber() {
        Random random = new Random();

        int[] lottoNumber = new int[6];

        for (int i = 0; i < lottoNumber.length; i++) {
           lottoNumber[i] = random.nextInt(42) + 1;
        }

        return lottoNumber;
    }
    // display player winning numbers
    public static void displayWinningNumbers(int[] playerNumber,int[] lottoNumber) {
        System.out.println("Winning numbers");
        boolean isMatch = false;

        for (int i = 0; i < lottoNumber.length; i++) {
            for (int j = 0; j < playerNumber.length; j++) {
                if (lottoNumber[i] == playerNumber[j]) {
                    System.out.print(playerNumber[j] + " ");
                    isMatch = true;
                }
            }
        }
```

```java
        if (!isMatch) {
            System.out.println("Not found!");
        }
        System.out.println();
    }
    public static void displayNumbers(int[] playerNumber, int[] lottoNumber) {
        System.out.println("Bet Numbers:");
        for (int i : playerNumber) {
            System.out.print(i + " ");
        }
        System.out.println();
        System.out.println("Generated Numbers:");
        for (int i : lottoNumber) {
            System.out.print(i + " ");
        }
        System.out.println();
    }
    // Display prize of the winners
    public static void displayPrize(int[] playerNumber,int[] lottoNumber) {
        int count = 0;
        double prize = 0;

        // Nested Loop
        for (int i = 0; i < lottoNumber.length; i++) {
            for (int j = 0; j < playerNumber.length; j++) {
                // Check Matching numbers
                if (lottoNumber[i] == playerNumber[j]) {
                    // if True count will add 1
                    count++;
                }
            }
        }
```

```java
        // Prize
        switch (count) {
            case 6 -> prize = 31621223.97;
            case 5 -> prize = 25000;
            case 4 -> prize = 1000;
            case 3 -> prize = 20;
            default -> {
            }
        }
        System.out.println("\nYou won: ₱ " + String.format("%,.2f",prize));
    }
}
```

Sample Output 1

```
--- Lotto ---
Enter number #1 : 25
Enter number #2 : 34
Enter number #3 : 12
Enter number #4 : R
Please enter a number!
Enter number #4 : 45
Number must be between 1 and 42!
Enter number #4 : 21
Enter number #5 : 12
Enter number #6 : 32

Bet Numbers:
25 34 12 21 12 32
Generated Numbers:
11 18 11 14 33 9
Winning numbers
Not found!


You won: ? 0.00
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
```

Methods

```java
import java.util.Scanner;
public class Mavenproject4
{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter your height (in cm): ");
        double height = scanner.nextDouble();
        System.out.print("Enter your weight (in kilograms): ");
        double weight = scanner.nextDouble();

        calculateBMI(height, weight);

    }
    // Calculate BMI
    public static void calculateBMI(double height, double weight) {
        double bmi = 0;
        String category = "";
        bmi = (weight / (height * height)) * 10000;

        if (bmi >= 30) {
            category = "Obesity";
        } else if (bmi > 25) {
            category = "Overweight";
        } else if (bmi > 18.5) {
            category = "Normal weight";
        } else {
            category = "Underweight";
        }
        System.out.printf("BMI: %.2f\n", bmi);
        System.out.println("Category: " + category);
    }
}
```

Sample Output 1

```
Enter your height (in cm): 120
Enter your weight (in kilograms): 50
BMI: 34.72
Category: Obesity
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  33.918 s
Finished at: 2026-05-02T13:37:48+08:00
------------------------------------------------------------------------
```

Sample Output 2

```
Enter your height (in cm): 200
Enter your weight (in kilograms): 55.5
BMI: 13.88
Category: Underweight
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  7.511 s
Finished at: 2026-05-02T13:38:58+08:00
------------------------------------------------------------------------
```

Arrays and Sorting

Maven.java × Main.java × Mavenproject4.java ×

Source History

```java
import java.util.Scanner;
import java.util.InputMismatchException;
public class Mavenproject4
{
        public static void main(String[] args) {
            Scanner scanner = new Scanner(System.in);
            System.out.print("Enter a number of integers: ");
            int size = scanner.nextInt();
            char choice = ' ';
            int[] arr = new int[size];

        for (int i = 0; i < arr.length; i++) {
            while (true) {
                try {
                System.out.print("Enter number " + (i + 1) + ": ");
                arr[i] = scanner.nextInt();
                break;
            } catch (InputMismatchException e) {
                System.out.println("Please enter a number!");
                scanner.next();
            }
            }
            }


            do {
                System.out.print("Enter a choice from the menu: ");
                choice = scanner.next().charAt(0);
                System.out.println();
                switch (choice) {
                    case '1':
                        displayNumbers(arr);
                        System.out.println();
                        break;
                    case '2':
                        displayEvenIndex(arr);
```

```java
                        System.out.println();
                        break;
                    case '3':
                        displayOddIndex(arr);
                        System.out.println();
                        break;
                    case '4':
                        displayAscOrder(arr);
                        System.out.println();
                        break;
                    case '5':
                        displayDescOrder(arr);
                        System.out.println();
                        break;
                }
            } while (choice != 'X');
    }
    public static void displayNumbers(int[] arr) {
        System.out.print("Numbers: ");
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
    public static void displayEvenIndex(int[] arr) {
        System.out.print("Values in even indexes: ");
        for (int i = 0; i < arr.length; i++) {
            if (i % 2 == 0) {
                System.out.print(arr[i] + " ");
            }
        }
        System.out.println();
    }
    public static void displayOddIndex(int[] arr) {
        System.out.print("Values in odd indexes: ");
        for (int i = 0; i < arr.length; i++) {
            if (i % 2 != 0) {
```

```java
                System.out.print(arr[i] + " ");
            }
        }
        System.out.println();
    }
    public static void displayAscOrder(int[] arr) {
        for (int i = 0; i < arr.length; i++) {
            for (int j = 0; j < arr.length - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.print("Values in ascending order: ");
        for (int i : arr) {
            System.out.print(i + " ");
        }
        System.out.println();
    }
    public static void displayDescOrder(int[] arr) {
        for (int i = 0; i < arr.length; i++) {
            for (int j = 0; j < arr.length - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.print("Values in descending order: ");
        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
```

Sample Output 1

```
--- exec:3.5.1:exec (default-cli) @ mavenproject4 ---
Enter a number of integers: 6
Enter number 1: 50
Enter number 2: 62
Enter number 3: 123
Enter number 4: 12354
Enter number 5: 3654
Enter number 6: 121
Enter a choice from the menu: 4

Values in ascending order: 50 62 121 123 3654 12354

Enter a choice from the menu: X
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  01:30 min
Finished at: 2026-05-02T13:48:38+08:00
------------------------------------------------------------------------
```

Stacks and Queues

```java
import java.util.Scanner;
public class Mavenproject4
{
        public static void main(String[] args) {
                Scanner scanner = new Scanner(System.in);
                String data = "";
                System.out.print("Enter any text: ");
                data = scanner.nextLine();

                char[] ch = data.toCharArray();
                int size = ch.length;

                char choice = ' ';
                do {
                    System.out.print("Enter a choice from the menu [Press X to exit the program]: ");
                    choice = scanner.next().charAt(0);

                    switch(choice) {
                        case 'A':
                            for (int i = 0; i < size; i++) {
                                System.out.print(ch[i]);
                            }
                            System.out.println();
                            break;
                        case 'B':
                            if (size == 0) {
                                System.out.println("Empty");
                            } else {
                                System.out.println("Character Popped");
                                size--;
                            }
                            break;
                        case 'C':
                            if (size == 0) {
                                System.out.println("Empty");
                            } else {
                                System.out.println("Top character is " + ch[size - 1]);
```

```
                    }
                    break;
                case 'D':
                    System.out.print("Enter character: ");
                    char letter = scanner.next().charAt(0);

                    char[] newLetter = new char[size + 1];
                    for (int i = 0; i < size; i++) {
                        newLetter[i] = ch[i];
                    }
                    newLetter[size] = letter;
                    ch = newLetter;
                    size++;
                    System.out.println("Character Pushed/Enqueued");
                    break;
                case 'E':
                    if (size == 0) {
                        System.out.println("Empty");
                    } else {
                        for (int i = 1; i < size; i++) {
                            ch[i - 1] = ch[i];
                        }
                        size--;
                        for (int i = 0; i < size; i++) {
                            System.out.print(ch[i]);
                        }
                    }
                    System.out.println();
                    break;
                case 'F':
                    if (size == 0) {
                        System.out.println("Empty");
                    } else {
                        System.out.println("Front Character is " + ch[0]);
                    }
                    break;
                case 'X':
                    System.out.println("End");
                    break;
            }
        } while (choice != 'X');
    }
}
```

Sample Output 1

```
Enter any text: PERSEVERE

A. Display the string.
B. Pop
C. Top
D. Pushed/Enqueued
E. Dequeue
F. Front

Enter a choice from the menu [Press X to exit the program]: A
PERSEVERE
Enter a choice from the menu [Press X to exit the program]: D
Enter character: S
Character Pushed/Enqueued
Enter a choice from the menu [Press X to exit the program]: A
PERSEVERES
Enter a choice from the menu [Press X to exit the program]: E
ERSEVERES
Enter a choice from the menu [Press X to exit the program]: F
Front Character is E
Enter a choice from the menu [Press X to exit the program]: B
Character Popped
Enter a choice from the menu [Press X to exit the program]: A
ERSEVERE
Enter a choice from the menu [Press X to exit the program]: X
End
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
```